

Assessment

You are required to create a Task Management Application with the following **requirement** to be implemented:

1) Backend (Flask)

- a) Implement CRUD APIs for tasks:
 - i) /tasks – GET all tasks
 - ii) /tasks – POST create a new task
 - iii) /tasks/<id> – PUT update task
 - iv) /tasks/<id> – DELETE task
 - v) Error handling and validation
- b) Authentication:
 - i) POST /auth/register – create a new user
 - ii) POST /auth/login – login user and return JWT
 - iii) Ensure only authenticated users can access tasks endpoint
- c) Database:
 - i) Store users and tasks in MySQL
 - ii) Include password hashing
 - iii) Proposed table structure for User and Task:

User	Task
Id	Id
Email	Title
Password_hash	Description
Role	Status
	User_id

2) Frontend (React.js)

- a) Login page for users
- b) Register page for new users
- c) Dashboard display tasks for the logged-in user
- d) Input field to add task
- e) Button to update and delete tasks
- f) Store JWT token securely and attach in API calls
- g) Handled unauthorized request gracefully (redirect to login)
- h) Admin can handle all tasks; regular users only manage their own tasks

3) Infrastructure

- a) Dockerfile for backend and frontend
- b) Docker-compose.yml integrating all services
- c) README with setup instruction and run test

4) OPTIONAL

- a) Cloud Deployment (Azure / AWS / GCP)

Deliverable Submission:

- Full backend and frontend code
- Docker-compose.yml to run all services locally
- README Instruction:
 - How to setup backend + frontend + database
 - How to run tests locally
 - (optional) cloud deployment